User Classes and Characteristics:

Customer: Users who will buy products from website.

Registered User: Users who will create account and login to website.

Normal User: User who will not create account.

Admin: Who will monitor all user related content like add product, remove product etc.

Developer: Users who will maintain and will develop the e-commerce website.

Operating Environment:

1. Node.js and Express.js will run on JavaScript runtime environment and Node Package Manager (npm)
2. At first, it will operate on Google Chrome and Microsoft Edge.
3. For the client-side, the operating system of end-user devices (Windows, Android, macOS, Linux).
4. We also use windows for server-site but it's good to use Linux for server-site.
5. Need CORS configurations to handle requests from different ports.
6. We will use MongoDB as a database for storing data in our project.
7. We will use Github for Distributing the data.

Constraints:

Technology Constraints: Use latest version of node.js , react and the npm packages we will install into our system .

Database Constraints: Use MongoDB atlas database because of it is "no sql" and it's a online version so that different collaborator can access this database easily .

Internet Constraints: Internet connection is always required while working on the system. One of the main reason of it as we use online MongoDB database.

Developer Constraints: Developer should need proper knowledge about React,Node.js .

Security Constraints: Secure our api so that other person use the api and can't get essential data from our system , also secure our router so that unauthorized user can't see web pages. When a user register account then his password can't directly write into database use npm package called "bycript" for hashing password.